

Structural Consistency Does Not Reliably Predict Retrieval Utility: An Exact-and-Heuristic Audit of Preference-Graph Repair for Multi-Ranker Retrieval

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Abstract

Purpose: Preference-graph repair is often treated as if restoring acyclicity improves ranking quality. Heuristic studies leave greedy graph-objective suboptimality open for null results, and construction is rarely experimental. This study tests whether heuristic or exact minimum-weight feedback-arc-set (MWFAS) edge-deletion repair yields a multiplicity-corrected retrieval gain for score-derived multi-ranker graphs.

Methods: On four public benchmarks, graphs were built from stored BM25, TF-IDF, and MiniLM scores under three vote-construction regimes. Unrepaired, greedy, and exact repairs were compared with paired sign-flip tests and Holm correction. Exact SCIP diagnosed greedy suboptimality for the stated edge-deletion objective.

Results: Construction dominated structure: a conservative regime was always acyclic; a permissive regime was cyclic for most queries. Normalization cut BM25 edge-weight share from **0.988** to **0.512**. Repair removed contradictory weight and changed top-*k* membership by **10.6%** when pools exceeded the cutoff. No repaired-versus-unrepaired nDCG comparison survived Holm correction in the canonical, larger-pool, or exact-repair families. Dataset-macro nDCG kept CombSUM competitive (**0.554**) with the best repaired hybrid (**0.546**). Exact repair removed less weight than greedy yet yielded no corrected retrieval gain.

Conclusion: Under this protocol, repair is structurally active but not a validated nDCG optimization. Missing Holm-corrected evidence is not equivalence;

smaller effects remain possible. The contribution is a factorized audit of construction, repair, extraction, and utility, with exact MWFAS only as an identification control.

Keywords: preference graphs, feedback arc set, graph repair, rank aggregation, information retrieval, retrieval evaluation

1 Introduction

Pairwise preferences are a natural form of ranking evidence: a judgment states only that one document should be ranked ahead of another for a given query. Such evidence arises from multiple ranking signals (lexical and dense retrievers, cross-encoders, or large language models) or from systems that elicit pairwise comparisons. Aggregating these judgments yields a query-specific weighted directed preference graph, where an edge $u \rightarrow v$ encodes evidence for ranking u ahead of v .

This representation preserves disagreement, but it is structurally fragile. Aggregating imperfectly correlated sources can produce directed cycles ($u \rightarrow v \rightarrow w \rightarrow u$), so that no total order is consistent with all retained edges. Restoring acyclicity by removing a minimum-weight set of conflicting edges is therefore a plausible preprocessing step before ranking extraction: a self-consistent graph seems a better input for any subsequent scoring rule.

That intuition conflates four stages that must be kept separate. *Construction* chooses which votes are retained and how they are weighted. *Structural repair* minimizes an internal graph objective: the weight of edges deleted to restore acyclicity. *Ranking extraction* turns the (possibly repaired) graph into an ordering. *Downstream evaluation* scores that ordering against external relevance judgments with a retrieval metric such as normalized discounted cumulative gain (nDCG). Repair optimizes agreement with the graph’s own edges; nDCG measures agreement with qrels. Nothing in the repair objective guarantees that better internal consistency improves retrieval utility (Section 3.5).

Existing evidence for a retrieval benefit is inconclusive for three reasons that prior heuristic-only or structure-only studies cannot close. First, most evaluations use heuristic repair, so a null retrieval result can be dismissed as heuristic suboptimality on the graph objective rather than as evidence about repair itself. Second, construction is rarely treated as an experimental factor, so repair effects cannot be separated from upstream choices that create inconsistency. Third, studies that report structural diagnostics alongside retrieval metrics rarely test the link under multiplicity-corrected paired inference; gains on the former are often treated as if they implied gains on the latter.

This paper is a controlled empirical and methodological audit of that assumption. It is not a new ranking algorithm, not a new MWFAS formulation, not a new exact solver, and not a state-of-the-art retrieval model. We treat construction as an experimental factor and compare unrepaired, greedy-repaired, and exactly repaired graphs across three vote-construction regimes on identical candidate sets and stored BM25,

TF-IDF, and MiniLM scores. Exact MWFAS repair, solved to solver-certified optimality with the open-source SCIP solver on the stated edge-deletion formulation, is an *identification control*: if certified optimal repair still yields no reliable retrieval gain of the detectable scale, greedy suboptimality with respect to that objective cannot explain the null. Empirically, construction choices decide whether repairable inconsistency exists at all. When the candidate pool exceeds the evaluation cutoff, repair changes top- k membership by an average of 10.6%. Yet no repaired-versus-unrepaired nDCG comparison survives Holm-corrected paired testing in any reported family, and simple graph-free fusion remains competitive with the best repaired graph method (Section 6).

The contribution is threefold and deliberately non-overlapping:

1. **A factorized preference-graph audit.** We separate construction, structural intervention, ranking extraction, and downstream utility as distinct empirical stages, report them with separate evidence, and apply paired, multiplicity-corrected inference to repaired-versus-unrepaired comparisons. The reusable pattern is this factorization plus an exact-repair diagnostic control (Sections 4, 6.5).
2. **Construction-sensitive structural evidence.** Under the studied regimes, construction choices determine whether cyclic structure exists; when it does, repair is a real structural intervention, including top- k membership change when $P > k$ and systematically lower removed weight under exact than under greedy repair.
3. **Protocol-conditional retrieval evidence.** No repaired-versus-unrepaired nDCG family survives Holm correction under this protocol, while graph-free fusion anchors remain competitive. Exact repair narrows the interpretation of that null by ruling out greedy suboptimality with respect to the specified MWFAS objective on certified instances, not by proving that repair never helps.

Scope of evidence.

Confirmatory evidence comes from multi-ranker retrieval on SciDocs, FiQA, HotpotQA, and BRIGHT with three base scorers and three vote-construction regimes; inference is query-level within reported Holm families. The 1,025 certified exact solves establish solver coverage, not 1,025 independent retrieval effects. A six-query LLM pilot is directional only. Results do not extend to cross-encoder or large-scale LLM rerankers, learned fusion, or interactive retrieval; they do not prove universal repaired–unrepaired equivalence; and they do not prove that repair is never beneficial outside this protocol.

2 Related Work

2.1 Pairwise Ranking and Preference Aggregation

Pairwise comparisons convert relative judgments into rankings along two traditions that answer different questions. *Probabilistic preference modeling* explains noisy outcomes through latent strengths, as in Bradley–Terry and Plackett–Luce models [4, 20, 29, 35], or recovers a ranking as the stationary distribution of a comparison

Markov chain [32]. Combinatorial Hodge theory separates a globally consistent component from a cyclic residual and therefore diagnoses inconsistency without treating cycle removal as a retrieval intervention [23, 48]. Learning methods such as GNNRank instead train digraph embeddings to reduce ranking upsets on pairwise-comparison graphs [17]. *Combinatorial aggregation and cycle repair*, by contrast, treat the observed preferences as fixed evidence and seek an ordering or graph edit that minimizes disagreement, from Kemeny consensus and web-scale aggregation rules [10, 25] to approximation algorithms linked to correlation clustering [1, 26]. Bias-aware pairwise ranking further shows that the comparison sampling process itself can introduce systematic bias before any repair step [13].

This paper follows the combinatorial tradition operationally—edge-deletion MWFAS repair—but asks a different empirical question from ranking-recovery benchmarks that optimize upset or disagreement objectives on comparison graphs [17, 47]: whether improving that graph-internal objective improves retrieval utility against external qrels (Section 3.4).

2.2 Pairwise and Graph-Based Ranking in Information Retrieval

In retrieval, pairwise and listwise comparisons are primarily tools for producing better rankings, not for auditing repair. Pairwise ranking prompting and related LLM paradigms ask whether using pairwise (or listwise) evidence improves effectiveness relative to pointwise scoring [37, 43]. Listwise and setwise systems reduce call cost while reordering candidates [15, 36, 53, 57], and tournament-style LLM ranking ensembles multiple group outcomes to mitigate input-order sensitivity [6]. Hybrid lexical/dense pipelines and graph-free fusion (RRF, CombSUM, Borda) remain strong alternatives that never build a preference graph [8, 14, 49, 55].

A smaller line builds an explicit preference graph and then aggregates it. PRP-Graph constructs a graph from LLM pairwise prompts and aggregates for text reranking [30]; LLM-RankFusion mitigates pairwise inconsistency across LLM rankers [54]. Those systems evaluate whether graph aggregation of pairwise LLM evidence improves ranking. The present study holds a score-derived multi-ranker graph fixed and asks whether *repairing* retained contradictions adds nDCG under corrected inference—a marginal-repair question rather than a pairwise-vs-pointwise bake-off [9, 44].

2.3 Inconsistency, Cycles, and Preference Repair

A directed preference graph admits a total order consistent with all retained edges if and only if it is acyclic; a DAG need not be transitively closed, and several linear extensions may remain. Aggregating disagreeing sources routinely creates cycles. Removing a minimum-weight edge set to restore acyclicity is the NP-hard MWFAS problem [24], addressed by greedy cycle heuristics [11], approximation schemes connected to correlation clustering [1, 26], and exact linear-ordering / branch-and-cut methods [2, 16]. Exact FAS optimization targets a graph objective; this paper uses

Table 1 Closest prior lines versus this manuscript’s factorized audit. “Corrected inference” means multiplicity-aware paired testing of repaired-versus-unrepaired retrieval effects.

Work	Setting	Inconsistency / exact	Diff. from this paper
PRP-Graph [30]	LLM pairwise prompts; graph; text reranking	Aggregate graph scores; no exact MWFAS	Tests pairwise graph aggregation, not repair vs. unrepaired under Holm
LLM-RankFusion [54]	Multi-LLM pairwise; graph; ranking fusion	Mitigate inconsistency before fusion; no exact MWFAS	Improves agreement/aggregation; no exact-MWFAS retrieval diagnostic
PGED [19]	Multi-evaluator prefs; graph; LLM eval/selection	Ensemble + denoise to DAG; no exact MWFAS	Downstream tasks are not document-retrieval nDCG with construction factors
TCR [28]	LLM-judge prefs; graph; RLAIF rewards	Greedy MAS / cycle filtering; no exact MWFAS	Optimizes RL stability, not qrel nDCG; preprint
TourRank [6]	LLM group ranking; no FAS graph; zero-shot ranking	Points ensemble vs. order bias; no exact MWFAS	No preference-graph repair stage
Earlier preprint [46]	Score-derived; graph; retrieval nDCG	Greedy repair only; no exact MWFAS	Same theme; superseded by exact+Holm protocol (Table 2)

certified exact solves only as an identification control for downstream retrieval interpretation, not as a new formulation or solver. Related combinatorial MWFAS ranking work likewise optimizes ranking-recovery metrics rather than retrieval nDCG under construction-factor audits [47].

LLM-as-judge settings exhibit the same cycles through positional and order effects [41, 56]. Mitigation methods may improve agreement or acyclicity and report gains on LLM evaluation, selection, or RLAIF objectives—for example preference-graph ensemble denoising [19] and recent cycle-breaking preference-graph rewards for reinforcement learning from AI feedback [28]. Those results support treating inconsistency as real, but they do not by themselves establish that repairing a document-retrieval preference graph improves qrel-based nDCG under multiplicity-corrected paired tests.

Table 1 summarizes the closest lines. Across them, either the task is not retrieval utility against qrels, or repair is absent, or exact MWFAS is not used as a control, or multiplicity-corrected repaired-versus-unrepaired inference is not the confirmatory design.

Table 2 records the relationship to the earlier preprint [46]: same four benchmarks and regime names, but greedy-only canonical repair and bootstrap intervals without the exact-SCIP diagnostic or Holm families used here. The present manuscript extends and re-analyzes partially overlapping evidence under a stricter protocol; it is not a resubmission of the preprint text, and the preprint’s numerical retrieval findings are superseded. Springer Nature journals encourage preprint posting and require DOI/license disclosure at submission; we disclose the preprint in the cover letter.

Table 2 Relationship to the author’s earlier Research Square preprint [46] (posted 2026-06-17; CC BY 4.0). The present manuscript is a distinct analysis that reuses the same public benchmarks and partially overlapping stored multi-ranker scores under a stricter evaluation protocol. Preprint-specific numerical retrieval claims are superseded here; the construction-sensitivity theme is retained and strengthened.

Dimension	Earlier preprint	This manuscript
Datasets	SciDocs, FiQA, HotpotQA, BRIGHT	Same four public benchmarks
Vote regimes	ms2/ms1/ms1_drop_mutual	Same regime names; updated construction and reporting protocol
Canonical repair	Greedy cycle-peeling for real-data claims	Greedy plus exact SCIP MWFAS as co-reported diagnostic
Exact MWFAS	Not used for canonical retrieval claims	Solver-certified optima on 1,025 instances
Inference	Paired bootstrap intervals on $\Delta nDCG$	Paired sign-flip tests with prespecified Holm families
Multiplicity	No family-wise Holm correction	Holm correction within confirmatory families
Main retrieval reading	Conditional / mixed dataset-dependent effects	No Holm-surviving repaired-vs-unrepaired nDCG gain
Claim status	Public preprint; not peer-reviewed	Supersedes preprint retrieval-effect claims under the present protocol

2.4 Structural Quality Versus Downstream Utility

Structural diagnostics (cyclicity, removed feedback-arc weight) are often reported without a corrected test against a downstream task metric. That pattern mirrors the data-quality distinction between *intrinsic* and *contextual* quality [50] and the broader lesson that upstream choices can reshape conclusions when only the final metric is inspected [33, 40]. Retrieval comparisons require multiplicity-aware paired testing [5, 21, 45], and non-significant results are not equivalence claims [27]. We found limited prior preference-graph and FAS ranking work that jointly (i) varies construction as a factor, (ii) compares unrepaired, greedy, and certified-exact repair, and (iii) tests repaired-versus-unrepaired retrieval utility under prespecified Holm families on multi-ranker document retrieval. The factorized audit in Section 1 targets that gap.

3 Background and Problem Formulation

This section formalizes the pipeline stages from Section 1. Instantiation details appear in Section 4.

3.1 Retrieval and Candidate Rankings

Let $q \in \mathcal{Q}$ denote a query drawn from a fixed query set, and let $D_q = \{d_1, \dots, d_n\}$ denote its query-specific candidate document set: the pool of documents that will be compared and ranked for q . Candidate sets are formed upstream of our pipeline,

typically from the union of documents scored by one or more base rankers, and are held fixed.

A base ranker $s \in S$ assigns a score $x_{q,s}(d) \in \mathbb{R}$ to each document $d \in D_q$ it evaluates. Scores may be missing for some documents. Because raw scores from different rankers are not directly comparable, they are normalized per query and per ranker before cross-ranker comparison. The normalization used in the canonical experiments is per-query, per-ranker min-max scaling,

$$\bar{x}_{q,s}(d) = \frac{x_{q,s}(d) - \min_{d' \in D_{q,s}} x_{q,s}(d')}{\max_{d' \in D_{q,s}} x_{q,s}(d') - \min_{d' \in D_{q,s}} x_{q,s}(d')}, \quad (1)$$

where $D_{q,s} \subseteq D_q$ is the subset of candidates scored by ranker s for query q . If all retained scores for a query-ranker pair are equal (range below 10^{-12}), the normalized vector is set to all zeros. Missing candidates are excluded from the range calculation and remain abstentions when votes are formed. Pairwise votes are built from normalized scores $\bar{x}_{q,s}$ (Section 3.2). The unnormalized construction is used only for a robustness ablation, as raw score margins are not a scale-neutral basis for comparing rankers.

Relevance judgments (qrels), which provide graded or binary labels for query-document pairs, are drawn from external human annotations and are not derived from ranker scores or preference graphs. All retrieval metrics are computed against these external judgments at a fixed cutoff k . For example, nDCG@ k scores only the top- k documents in a ranking. The relationship between the candidate pool size $P = |D_q|$ and the cutoff k is an explicit design choice (Section 4). Repair can change the top- k set only when $P > k$.

3.2 Derived Weighted Preference Graph

For each query q , the normalized ranker scores $\bar{x}_{q,s}$ are converted into a weighted directed preference graph

$$G_q = (V_q, E_q, w_q), \quad V_q = D_q, \quad (2)$$

where a directed edge $(u, v) \in E_q$ with weight $w_q(u, v) > 0$ represents the strength of evidence for ranking document u ahead of document v . The graph is a derived object, not ground truth. Its structure (edges, directions, and weights) is determined by an explicit construction procedure applied to the normalized scores, not by intrinsic properties of the documents.

Construction proceeds ranker by ranker and pair by pair. For each ranker s and each scored document pair $\{u, v\} \subseteq D_q$, a directional vote ($u \succ v$ or $v \succ u$) is cast based on the normalized scores. The vote strength is derived from the normalized score margin, $|\bar{x}_{q,s}(u) - \bar{x}_{q,s}(v)|$. Ties in normalized score or pairs where a document is unscored result in an abstention. These are treated as an absence of evidence, not evidence of indifference. Votes are aggregated per directed pair. A retention rule, the *vote-construction regime*, then determines which pairs are retained as edges in G_q .

This study evaluates three vote-construction regimes, varying only the retention rule while holding the candidate set and ranker scores fixed:

- **ms2** (conservative): retain a direction only with majority support from rankers and an aggregated margin above a threshold. This conservative regime suppresses most two-way contradictions and typically produces acyclic graphs.
- **ms1** (permissive): retain every ranker-supported directional vote. This permissive regime allows for direct mutual contradictions—where both (u, v) and (v, u) are retained—and thus longer cycles.
- **ms1_drop_mutual**: similar to **ms1**, but drops pairs with bidirectional support during vote construction. This isolates cyclicity arising from longer-cycle nontransitivity rather than direct mutual contradictions. It is distinct from a post-hoc modification of an **ms1** graph.

These regimes are not proposed as optimal construction rules but serve as a controlled sweep to establish graph construction as a first-class experimental factor. Section 4 reports its strong effect on graph cyclicity. Since the vote-construction regime determines the level of inconsistency, any observed repair effect must be interpreted in the context of the regime that produced the graph.

3.3 Cycles and Structural Inconsistency

A weighted directed graph G_q is *acyclic* if it contains no directed cycle $u_1 \rightarrow u_2 \rightarrow \dots \rightarrow u_m \rightarrow u_1$; equivalently, its strongly connected components are all singletons. An acyclic graph G_q admits at least one consistent linear extension. Conversely, a cyclic graph cannot be represented by any single total order over D_q without violating at least one edge.

We report two specific measures of structural inconsistency rather than a single omnibus cyclicity score. The first is the cyclic-query rate: the fraction of queries for which G_q is cyclic under a given construction regime. The second measure distinguishes cyclicity from direct mutual contradictions (i.e., pairs retained in both directions) from that which persists after their removal. We report two related diagnostics: post-aggregation deletion of reciprocal edges from an **ms1** graph, and the separate **ms1_drop_mutual** construction. These are not guaranteed to coincide because they handle mutual contradictions at different stages and may use different thresholds (Section 5.1). Conflating them would misrepresent the source of cyclicity.

Acyclicity is a purely structural property. It guarantees the existence of a consistent total order but does not guarantee the correctness of the retained edges, agreement with relevance judgments, or superior performance of any particular ranking extraction rule. We distinguish between transitivity, acyclicity, evidential correctness, and retrieval optimality. Accordingly, Sections 3.4 and 3.5 separate the repair objective (feedback-arc weight) from the evaluation metric (nDCG).

3.4 Minimum-Weight Feedback-Arc-Set Repair

Given a preference graph $G_q = (V_q, E_q, w_q)$, repair seeks a minimum-weight feedback arc set $F_q^* \subseteq E_q$ whose removal renders the graph acyclic:

$$F_q^* = \arg \min_{F \subseteq E_q} \sum_{e \in F} w_q(e) \quad \text{s.t.} \quad (V_q, E_q \setminus F) \text{ is acyclic.} \quad (3)$$

This is the minimum-weight feedback arc set (MWFAS) problem (Section 2.3); the repaired graph is $\tilde{G}_q = (V_q, E_q \setminus F_q^*, w_q)$. Equation (3) is purely structural: it minimizes disagreement with the graph’s own edges and does not mention relevance judgments or nDCG.

We evaluate two co-equal solvers for Equation (3). **Greedy edge-deletion repair** iteratively finds some directed cycle (NetworkX DFS order) and deletes that cycle’s minimum-weight edge (weight ties broken by that cycle’s edge order), repeating until acyclic. It is deterministic on a fixed graph but not minimum-weight; on cyclic queries here it typically *over-removes* total weight relative to a certified optimum (Section 5.4). **Exact edge-deletion repair** solves Equation (3) with the linear-ordering MIP below (PySCIPOpt/SCIP [2, 16]). Let $\text{before}[u, v] \in \{0, 1\}$ mean u precedes v in a total order of V_q :

$$\begin{aligned} \min \quad & \sum_{(u,v) \in E_q} w_q(u, v) \text{before}[v, u] \\ \text{s.t.} \quad & \text{before}[u, v] + \text{before}[v, u] = 1 \quad \forall u \neq v, \\ & \text{before}[a, b] + \text{before}[b, c] + \text{before}[c, a] \leq 2 \quad \forall \text{distinct } a, b, c, \\ & \text{before}[u, v] \in \{0, 1\} \quad \forall u \neq v. \end{aligned} \quad (4)$$

Edge (u, v) is removed iff $\text{before}[v, u] = 1$. Reciprocal pairs may both contribute and both be removed. The optimum equals Equation (3). Graphs enter the exact-repair evidence only with SCIP proven-optimal status (relative gap 0.0).

Exact repair is a *diagnostic control* for greedy *suboptimality w.r.t. Equation (3)* on certified instances: no feasible edge deletion restores acyclicity with less weight under that objective. Exactness does not extend to retrieval utility, relevance-aware repair, edge reversal, soft aggregation, or other interventions. No scalability claim is made beyond the sizes studied here.

3.5 Ranking Extraction and Evaluation

A final ranking is extracted from the unrepaired graph G_q , the repaired graph $\tilde{G}_q$, or the upstream ranker signals alone. The canonical experiments compare four families of extraction rules. Graph-dependent rules are evaluated on both unrepaired and repaired graphs:

- **Prior-only**: ranks by an upstream per-document score, independent of the graph.

- **Graph-free fusion:** methods such as reciprocal rank fusion, CombSUM, and Borda-style fusion, which operate on the constituent rankers’ outputs without constructing a preference graph.
- **Graph-only extraction:** a Copeland-style score $s_{\text{cop}}(u) = \deg^+(u) - \deg^-(u)$ (unweighted out-degree minus in-degree); a weighted balance score $s_{\text{bal}}(u) = \sum_{(u,v) \in E} w(u,v) - \sum_{(v,u) \in E} w(v,u)$ (weighted out-edge mass minus in-edge mass); and one or more stationary-distribution scores computed from a Markov chain over the (weighted) graph, in the spirit of Rank Centrality (Section 2.1). Section 4 details the specific instantiations of these rules, which include multiple Markov-chain variants and other centrality measures. Each rule is applied to both the unrepaired (G_q) and repaired ($\tilde{G}_q$) graphs, holding the rule constant to isolate the effect of repair.
- **Hybrid prior-plus-graph extraction:** the graph score is combined with a graph-free prior score,

$$h_q(d) = \hat{s}_q^{\text{prior}}(d) + \alpha \hat{s}_q^{\text{graph}}(d), \quad (5)$$

where both score components are independently min-max normalized per query. The primary analysis uses $\alpha = 0.3$, with other values tested for robustness. The prior $\hat{s}_q^{\text{prior}}$ is reciprocal rank fusion over the three base rankers on D_q with $k=60$, not a simple score sum; final ties break by document id. The method is unrepaired when $\hat{s}_q^{\text{graph}}$ is computed from G_q and repaired when computed from $\tilde{G}_q$.

Retrieval effectiveness is measured with normalized discounted cumulative gain (nDCG) [22]: for relevance labels $\text{rel}_1, \dots, \text{rel}_k$ at ranks $1, \dots, k$,

$$\text{DCG}@k = \sum_{i=1}^k \frac{2^{\text{rel}_i} - 1}{\log_2(i + 1)}, \quad \text{nDCG}@k = \frac{\text{DCG}@k}{\text{IDCG}@k}, \quad (6)$$

where IDCG@ k is the ideal DCG. We compute nDCG per query and compare unrepaired and repaired variants of each extraction rule with all other factors held fixed (Section 4). Because Equation (3) uses only graph-internal weights while nDCG uses external qrels, optimizing repair is not the same problem as improving nDCG; Sections 4 and 5 test whether the two are linked empirically under this protocol.

4 Methodology

This section instantiates Section 3: research questions, pipeline, datasets, rankers, construction, repair, extraction, baselines, metrics, and statistical protocol for the evidence in Section 5. It documents what was measured and how, not outcomes.

4.1 Research Questions

The study answers four questions that operationalize the separation of construction, structural repair, and retrieval utility (Section 1):

RQ1 How much structural inconsistency – cyclicity and contradictory edge mass – is present in derived multi-ranker preference graphs, and how strongly does it depend on graph-construction choices (Section 3.2) rather than on the underlying documents?

RQ2 How do greedy and exact minimum-weight feedback-arc-set repair (Section 3.4) differ in the structural change they produce – removed edge weight, the chosen feedback arc set, and top- k membership when the candidate pool exceeds the evaluation cutoff – on the same graphs?

RQ3 Does preference-graph repair, heuristic or exact, produce a statistically reliable improvement in downstream retrieval effectiveness (nDCG) relative to the unrepaired graph, once paired inference and multiplicity correction are applied – and how do repaired graph methods compare with simple graph-free fusion anchors?

RQ4 Is any observed retrieval null in RQ3 attributable to greedy-repair suboptimality rather than to repair itself, once exact MWFAS is used as a diagnostic control under matched construction, pooling, and extraction choices?

These questions concern the marginal effect of repair under this protocol, not a learned repair policy or a retrieval leaderboard. Fairness for RQ3–RQ4 means holding pool, scores, construction, extraction, and cutoff fixed while varying only the repair branch (Section 4.8).

4.2 Experimental Pipeline

Every reported result is produced by the same nine-stage pipeline, run once per query under each combination of dataset, vote-construction regime, and evaluation cell studied:

1. obtain stored per-document scores from each base ranker (Section 4.4) for a fixed candidate pool;
2. normalize scores per query and per ranker (Equation (1));
3. derive weighted pairwise votes from normalized score margins (Section 3.2);
4. construct the query-specific weighted directed preference graph G_q under one vote-construction regime (Section 4.5);
5. extract unrepaired rankings directly from G_q and from the prior score alone;
6. repair G_q into $\tilde{G}_q$ with greedy cycle-peeling or exact SCIP-based MWFAS solving (Section 4.6);
7. extract repaired rankings from $\tilde{G}_q$ using the same extraction rule applied to G_q (Section 4.7);
8. evaluate every extracted ranking against relevance judgments with nDCG at a prespecified cutoff (Section 4.9); and
9. conduct paired, multiplicity-corrected statistical inference on repaired-versus-unrepaired differences (Section 4.10).

Stages 1–4 are shared by every downstream comparison; stages 5–9 are repeated once per extraction rule and once per repair method (unrepaired, greedy, exact) being compared. No stage regenerates upstream retrieval: stage 1 reuses fixed, versioned score files, so every reported statistic is reproducible from stored intermediates without repeating retrieval or querying a paid API (Section 4.11).

Figure 1 summarizes the pipeline: greedy and exact repair are parallel branches into shared extraction and evaluation, so differences are attributable to the repair step alone.

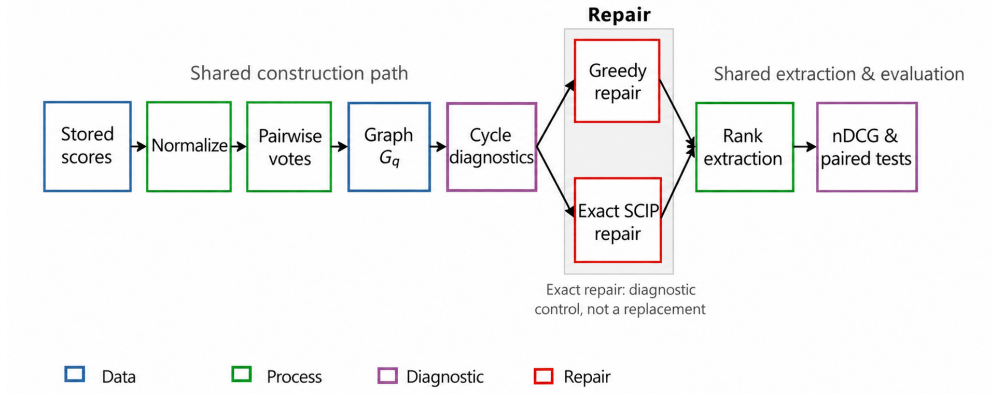


Fig. 1 Experimental pipeline answering RQ1–RQ4. Shared stages 1–4 build a preference graph from fixed multi-ranker scores; stages 5–9 then evaluate unrepaired, greedy-repaired, and exact-repaired branches with identical extraction and nDCG cutoffs, so retrieval comparisons attribute differences to the repair step alone.

4.3 Datasets and Retrieval Tasks

The study uses four public retrieval benchmarks chosen for domain diversity under a shared offline reranking setting, not as coverage of all retrieval workloads: SciDocs [7] (scientific document / citation-related retrieval), FiQA [31] (financial question-answering), HotpotQA [52] (multi-hop Wikipedia QA), and BRIGHT [42] (public examples configuration; reasoning-oriented retrieval). SciDocs, FiQA, and HotpotQA also appear in BEIR-style zero-shot benchmarking [44]. Together they stress construction and repair across scientific, financial, multi-hop, and reasoning tasks with public qrels; they do not claim generality to web-scale search, conversational retrieval, or interactive sessions.

Base-ranker scores, query IDs, and relevance judgments are reused from fixed stored files; no upstream retrieval is regenerated. The evaluation setting is candidate-pool reranking over previously retrieved documents rather than a full collection scan. Queries need at least one qrels-eligible relevant document among candidates scored by at least one ranker; this excludes 18 of HotpotQA’s 70 stored queries and leaves every other dataset’s stored queries usable. Candidate pools are the union of documents scored by at least one base ranker, truncated by the primary pool policy (Section 4.5); Table 3 reports depths and evaluated (P, k) cells. Usable query counts (50–120 per dataset) are modest by large-track standards, which is why power and minimum-detectable-effect reporting accompany the confirmatory nulls (Section 4.10).

4.4 Base Rankers and Judgment Sources

Three base rankers supply heterogeneous scores that become pairwise votes. Their role is a controlled multi-source substrate for construction and repair, not the strongest

Table 3 Datasets and evaluation cells for the offline reranking study. Usable: queries retaining at least one relevant candidate after qrels filtering. Depth: common complete stored score depth across BM25, TF-IDF, and MiniLM. Canonical cells set pool size P equal to metric cutoff k ; larger-pool cells set $P > k$ so repair can change top- k membership (RQ2–RQ3).

Dataset	Usable	Depth	Evaluated (P, k) cells
SciDocs	120	50	canonical (20, 20); larger-pool (20, 5), (20, 10), (20, 20), (50, 5), (50, 10), (50, 20)
FiQA	120	50	same as SciDocs
HotpotQA	52	35	canonical (10, 10); larger-pool (10, 5), (35, 5), (35, 10), (35, 20)
BRIGHT	50	50	same as SciDocs

available retrieval stack. **BM25** [38] (Okapi BM25, $k_1 = 1.5$, $b = 0.75$) and **TF-IDF** [39] (cosine logarithmic TF-IDF) are classical lexical signals; **MiniLM** [51] (**all-MiniLM-L6-v2** cosine embeddings, not fine-tuned here) is a fixed small dense signal. They are deliberately transparent and weaker than cross-encoders or large LLM rerankers so that any repair effect can be isolated from upstream ranker changes. All three score the same candidate pool; higher scores indicate stronger estimated relevance before normalization (Equation (1)).

4.5 Preference-Graph Construction

The primary construction protocol normalizes every ranker’s scores with per-query, per-ranker min-max scaling (Equation (1)) and extracts directional votes from normalized score margins, following Section 3.2. Vote retention uses a retention-matched thresholding rule that never consults qrels or nDCG. On each dataset, a raw-margin (unnormalized) construction with the same support rule supplies targets: per-ranker retained-vote rates and the regime’s retained directed edge count. For the normalized protocol, each ranker’s vote-margin threshold is the empirical quantile of that ranker’s pooled pairwise normalized margins matching the raw retained-vote rate ($q = 1 - \text{target_rate}$). Given those vote thresholds, the regime-level aggregate weight threshold is chosen by a discrete search over candidate aggregate weights to minimize the absolute gap in retained edge count versus the raw baseline for that regime (ties break toward the smaller threshold). Vote-margin thresholds are therefore per dataset and per ranker, shared across regimes within a dataset; aggregate thresholds are per dataset and per regime; neither is per query. Thresholds are fixed from the full usable query set of that dataset before retrieval metrics are computed. This is unsupervised collection-level protocol calibration: it uses only stored ranker scores, never qrels or nDCG, so it is not relevance-label leakage, but it is transductive with respect to the evaluated query list (no held-out calibration split; Section 7.1). Matching is approximate (minimum absolute edge-count gap), not exact equality of every rate. Canonical values appear in Table 4. Support requirements are fixed per regime: **ms2** requires support from at least two of the three rankers; **ms1** and **ms1_drop_mutual** require support from at least one ranker, differing in whether contested (two-way) pairs are dropped at vote construction (Section 3.2).

Table 4 Retention-matched thresholds for the primary protocol (`primary_minmax_retention_matched`). Vote thresholds are shared across regimes within a dataset; aggregate thresholds and drop-mutual flags are regime-specific. Matching target: raw-margin retained-vote rates (per ranker) and raw retained edge count (per regime). Calibration: unsupervised collection-level protocol thresholds on the same usable queries later evaluated; qrels/nDCG are not used (transductive query list; Section 7.1). Regime abbreviations: `ms1`, `ms1dm=ms1_drop_mutual`, `ms2`.

Dataset	Reg.	BM25	MiniLM	TF-IDF	Agg.	Sup.	Drop
SciDocs	<code>ms1</code>	0.00468	0.274	0.340	0.0108	1	no
SciDocs	<code>ms1dm</code>	0.00468	0.274	0.340	0	1	yes
SciDocs	<code>ms2</code>	0.00468	0.274	0.340	0	2	no
FiQA	<code>ms1</code>	0.00622	0.276	0.375	0.0120	1	no
FiQA	<code>ms1dm</code>	0.00622	0.276	0.375	0	1	yes
FiQA	<code>ms2</code>	0.00622	0.276	0.375	0	2	no
HotpotQA	<code>ms1</code>	0.00317	0.272	0.228	0	1	no
HotpotQA	<code>ms1dm</code>	0.00317	0.272	0.228	0.00341	1	yes
HotpotQA	<code>ms2</code>	0.00317	0.272	0.228	0.295	2	no
BRIGHT	<code>ms1</code>	0.00108	0.240	0.555	0.0114	1	no
BRIGHT	<code>ms1dm</code>	0.00108	0.240	0.555	0.0119	1	yes
BRIGHT	<code>ms2</code>	0.00108	0.240	0.555	0	2	no

Candidate pools use one primary policy: for each query, the top- P documents of a reciprocal-rank-fusion ranking computed over the union of the three rankers’ native (unnormalized) score lists. This “RRF-centered” pool policy is one specific, named choice among several equally valid ways to truncate a candidate pool, and it determines which documents are eligible to appear in D_q at all – a decision made once, upstream of score normalization and graph construction. Four alternative, independently defined pool policies (equal per-ranker depth, round-robin interleaving across rankers, a single-ranker pool, and a CombSUM-ranked pool) are evaluated as a robustness check (Section 5) to test whether the study’s structural and retrieval conclusions depend on this specific choice.

Algorithm 1 summarizes graph construction for a single query, reflecting the procedure implemented in the repository’s graph- and vote-construction code. The description is intentionally compact: the contribution is not a new preference extractor, but a controlled study of how these construction choices affect cyclicity, repair, and downstream retrieval.

Algorithm 1 Preference-graph construction for one query

Require: Query q ; candidate set D_q ; base rankers $S = \{\text{BM25}, \text{TF-IDF}, \text{MiniLM}\}$ with scores $x_{q,s}$; construction regime $r \in \{\text{ms2}, \text{ms1}, \text{ms1_drop_mutual}\}$

Ensure: Weighted directed preference graph $G_q = (V_q, E_q, w_q)$

- 1: $V_q \leftarrow D_q$; initialize an empty weighted edge accumulator
- 2: **for** each ranker $s \in S$ **do**
- 3: Normalize $x_{q,s}$ to $\bar{x}_{q,s}$ (Equation (1))
- 4: **for** each unordered candidate pair $\{u, v\} \subseteq D_q$ scored by s **do**
- 5: **if** $|\bar{x}_{q,s}(u) - \bar{x}_{q,s}(v)|$ is below the vote-margin threshold, or s does not score both u and v **then**
- 6: Abstain on $\{u, v\}$ for ranker s
- 7: **else**
- 8: Accumulate directional support for $u \succ v$ or $v \succ u$ (whichever $\bar{x}_{q,s}$ supports), weighted by the normalized margin
- 9: **end if**
- 10: **end for**
- 11: **end for**
- 12: Apply regime r 's support rule and aggregate weight threshold to every accumulated directed pair
- 13: **if** $r = \text{ms1_drop_mutual}$ **then**
- 14: Remove every unordered pair retained in both directions
- 15: **end if**
- 16: $E_q \leftarrow$ every retained directed pair with positive aggregated weight
- 17: **return** $G_q = (V_q, E_q, w_q)$

4.6 Repair Methods

Two repair methods solve Equation (3), reported as co-equal throughout; their roles are defined in Section 3.4. **Greedy repair** is the repository's deterministic cycle-peeling edge-deletion heuristic described in Section 3.4 (not the classical Eades–Lin–Smyth linear-arrangement heuristic [11]). It deletes edges until the graph is acyclic; it does not claim a minimum-weight feedback arc set. **Exact repair** solves Equation (4) with open-source SCIP via PySCIPOpt (pinned version 6.2.1), requiring proven-optimal status (relative gap 0.0) and a 300-second per-query safety time limit that was never approached at the sizes studied here. The formulation was cross-checked against brute-force enumeration on synthetic and real query graphs. Exact repair remains the diagnostic control of Section 3.4 (RQ4; Section 5.4).

4.7 Ranking Extraction Methods

Extraction methods fall into four groups, extending Section 3.5's general definitions with the specific methods this study evaluates and stating precisely which methods enter which comparison family (Section 4.8).

Prior-only and graph-free fusion rank candidates without constructing or consulting a preference graph: the prior-only method ranks by the same RRF prior used

Table 5 Extraction-method implementation parameters matching the canonical pipeline. Direction: edge $u \rightarrow v$ means u preferred over v . Final ranking sorts by descending score then ascending document id unless noted.

Method	Main parameter(s) / ties	Implementation note
Copeland	$\text{deg}^+ - \text{deg}^-$; id tie-break	Isolates score 0
Balance	weighted out-in; id tie-break	Uses edge weights
Markov (damped)	restart 0.15; tol 10^{-10} ; 10^4 iter	Loser $\rightarrow$ winner + uniform restart
Rank Centrality	self-loops; tol 10^{-8} ; 200 iter	No teleport; normalize π
PageRank	damping 0.85; tol 10^{-6} ; 100 iter	Reversed graph; weighted
Bradley-Terry	MM; tol 10^{-6} ; 100 iter	Uniform init; isolates 0
Topological	NetworkX <code>topological_sort</code>	DAG-only; construction-order ties
Priority-topo	RRF prior ($k=60$) among sources	Higher prior, then larger id
Hybrids	$\alpha=0.3$; min-max each side	$h = \delta^{\text{prior}} + \alpha \delta^{\text{graph}}$; const. $\rightarrow 0$

in Equation (5); reciprocal rank fusion, CombSUM, and Borda-style fusion combine the three rankers’ individual per-query rankings or scores directly [8, 14].

Graph-only extraction computes a score from G_q or $\tilde{G}_q$ alone. Table 5 records the implementation parameters used throughout. The Copeland score s_{cop} and balance score s_{bal} (Section 3.5) are evaluated on every dataset, regime, and repair method. Two related but distinct Markov-chain scores are also evaluated: a damped stationary-distribution score with uniform restart (“Markov”), and an undamped Rank Centrality chain with explicit self-loops [32]. Standard graph PageRank on the reversed preference graph [34], Bradley-Terry MM MLE with edge weights as win counts [4, 20], and (on repaired DAGs only) NetworkX topological order plus prior-priority topological order complete the set.

Hybrid prior-plus-graph extraction combines a graph-only score with the RRF prior via Equation (5). Copeland and balance hybrids use $\alpha = 0.3$ in the primary analysis, with $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$ as a robustness check; a Markov hybrid at the same fixed α enters the expanded family (Section 4.8).

As in Section 3.5, every graph-dependent rule is evaluated on both G_q and $\tilde{G}_q$ wherever both are defined. Evaluating graph-only rules alongside hybrids, and reporting the α -sweep, further addresses the concern that a fixed hybrid mixing weight could mask a genuine repair effect at $\alpha = 0.3$.

4.8 Baselines and Comparisons

Table 6 assigns every method a comparison role. Score sources (Section 4.4) build the graph and are not retrieval competitors in the repair contrast. Graph-free fusion anchors (prior-only, RRF, CombSUM, Borda) never construct a preference graph and ask whether graph methods earn their complexity. Unrepaired graph extractors are the matched counterfactual; repaired extractors (greedy or exact) are the intervention under test for RQ3–RQ4. Two overlapping method families are reported separately

Table 6 Controlled comparison set for isolating graph construction and repair (not a retrieval leaderboard). Family: primary = greedy repaired-vs-unrepaired; expanded = exact repaired-vs-unrepaired; struct. = exact-vs-greedy only. Repair variants: U = unrepaired, G = greedy, E = exact.

Method	Family	Graph?	Prior?	Repair variants compared
Prior-only	primary, expanded	no	yes	n/a (no graph)
RRF	primary, expanded	no	no	n/a (no graph)
CombSUM	primary, expanded	no	no	n/a (no graph)
Borda fusion	primary, expanded	no	no	n/a (no graph)
Copeland	primary	yes	no	U, G, E
Balance	primary	yes	no	U, G, E
Markov (damped)	primary	yes	no	U, G, E
Copeland hybrid	primary	yes	yes ($\alpha=0.3$)	U, G, E
Balance hybrid	primary	yes	yes ($\alpha=0.3$)	U, G, E
PageRank	expanded	yes	no	U, E
Rank Centrality	expanded	yes	no	U, E
Markov hybrid	expanded	yes	yes ($\alpha=0.3$)	U, E
Bradley–Terry	expanded	yes	no	U, E
Topological	struct. only	yes	no	G vs. E only
Priority-topological	struct. only	yes	yes (tie-break)	G vs. E only

rather than pooled: a *primary family* (four graph-free anchors; Copeland, balance, and Markov as graph-only and hybrid rules) backs greedy repaired-versus-unrepaired comparisons; an *expanded family* adds PageRank, Rank Centrality, a Markov hybrid, and Bradley–Terry to the exact-repair diagnostic so that RQ4 is not limited to the primary extractors. Topological and priority-topological ordering appear only in exact-versus-greedy checks, because they require an already acyclic graph.

Methods omitted from Table 6 fall into three classes (Section 7.3): stronger modern retrieval baselines that answer a leaderboard question; neural graph rankers trained to minimize upsets rather than test marginal repair on a fixed score-derived graph; and systems that never produce a preference graph and therefore cannot enter the unrepaired-versus-repaired contrast. Their absence is an external-validity limitation, not an implication that the included anchors are state of the art.

4.9 Evaluation Metrics

The primary retrieval metric is normalized discounted cumulative gain (nDCG) at a prespecified cutoff k (Equation (6)), evaluated per query and averaged across a dataset’s usable queries; higher values indicate better agreement with graded relevance judgments. Mean reciprocal rank (MRR) and mean average precision (MAP) are

Table 7 Prespecified Holm families (claim-to-inference map). Confirmatory rows support primary nulls; robustness rows do not replace them. Counts match Tables 8 and 10.

Family	Question	Scope	Methods	n
Active canonical	Greedy repair gain?	$4 \times \text{ms1}$, $P=k$	Prim. 5 pairs	20
Full canonical	All regimes?	4×3 regimes, $P=k$	Prim. 5 pairs	60
Active larger-pool	When $P > k$?	ms1 , larger-pool	Prim. 5 pairs	110
Exact-repair canonical	Exact vs unrepaired?	$4 \times \text{ms1}$, $P=k$	Exp. 9 pairs	36
Exact-repair larger-pool	Exact under $P > k$?	larger-pool cells	Exp. 9 pairs	56
Exact vs greedy	Exact vs greedy ret.?	pooled / per-cell	Prim.+topo	35/399
Alternative-pool	Pool-policy artifact?	alt. pools	vs RRF/CombSUM	8
Equivalence (TOST)	Narrow equivalence?	active larger-pool	Prim. pairs	110

computed as secondary diagnostics only and are never used as a primary confirmatory metric. All three are computed at the same evaluated (P, k) cell (Table 3) for every method and repair variant being compared.

Following Section 3.5, the repair objective, structural diagnostics, and retrieval metrics are reported separately and never treated as interchangeable proxies.

4.10 Statistical Analysis

The unit of statistical inference is the query: repaired-minus-unrepaired nDCG once per query. Paired evaluation is required because methods share queries and pools [45]. Primary greedy confirmatory families (Table 8) use a two-sided Monte Carlo paired sign-flip test on the mean difference (10,000 sign patterns, seed 17, add-one p), matching the canonical `full_calibrated_core` / pool-cutoff pipelines; zeros are flip-invariant. Holm step-down [18] at $\alpha = 0.05$ controls family-wise error within each prespecified family (Table 7) [5, 21], but does not model hierarchical dependence when the same queries recur across regimes, pools, or extractors.

Exact-repair, power/MDE, and related analyses use `statistical_inference`: the same two-sided sign-flip null, with exact enumeration when nonzero differences $m \leq 20$ else Monte Carlo (10,000, seed 17), plus percentile/basic/BCa bootstrap 95% intervals (10,000 resamples, seed 13) [12]. Intervals, leave-one-out influence, BH FDR alongside Holm [3], and 80%-power MDE/TOST at prespecified margins [27] characterize detectable scale and limited equivalence; they do not replace confirmatory Holm or convert nonsignificance into blanket equivalence (Section 2.4).

For the appendix-scoped real-large-language-model pilot (Section 1), the unit of inference is the query *cluster*: six queries, each replicated across several provider, paradigm, and repetition conditions, are treated as six independent clusters, never as row-level independent observations. Row-level analysis of this pilot’s replicated rows (correct unit: $n = 6$; incorrect unit: one row per replicate) systematically overstates the pilot’s evidential weight and is not used anywhere in this study.

4.11 Reproducibility and Implementation

The full pipeline (Section 4.2) is implemented in Python, using `networkx` for graph representation and cycle detection, `numpy/scipy` for numerical and statistical routines, and `PySCIP0pt` 6.2.1 (open-source SCIP) for exact repair (Section 4.6); the MiniLM ranker (Section 4.4) uses the `sentence-transformers` library to load the fixed, pretrained `all-MiniLM-L6-v2` checkpoint. Every Monte Carlo or bootstrap procedure (Section 4.10) uses a fixed random seed, so that reported p -values and intervals are exactly reproducible from the same stored inputs. No stage of the pipeline used to produce a main-paper statistic requires a paid API call: base rankers use fixed, versioned, locally computed or stored score files, and the appendix-scoped real-LLM pilot (which does call paid provider APIs) is never a source of a main-paper number. All code, stored intermediates, and the scripts used to regenerate every reported statistic and table are available at the repository referenced in Section 9.

Generative-AI assistance. Anthropic Claude was used for software-development assistance, analysis and audit utilities, figure preparation, and drafting and revising portions of the manuscript. The author independently reviewed and verified all AI-assisted code, analyses, figures, citations, and text against the source materials and underlying evidence and assumes full responsibility for the work. No generative-AI tool is credited as an author.

5 Results

Results follow RQ1–RQ4 (Section 4.1). Each subsection states the scientific question it answers and reports only the corresponding structural or retrieval evidence. Unless noted, statistics use the canonical `primary_minmax_retention_matched` protocol (Section 4.5). Significance statements use the Holm-corrected decision at $\alpha = 0.05$ in the relevant family (Section 4.10); raw p -values appear only beside the corrected decision.

5.1 Structural Inconsistency Before Repair

Question (RQ1).

How much cyclicity depends on vote construction rather than documents?

The vote-construction regime, not the underlying documents, determines whether a query’s preference graph is cyclic at all. Figure 2 reports, for every dataset and regime, the fraction of queries whose graph contains a directed cycle, that same fraction after direct mutual contradictions are removed, and the mean feedback-arc weight repair would need to remove, expressed as a fraction of total graph weight, together with the active-regime before/after comparison.

The conservative regime, `ms2`, is acyclic on every query in every dataset in the observed evidence (Section 3.2): repair has no graph to act on under this regime. The permissive regime, `ms1`, is cyclic on the large majority of queries in every dataset, ranging from 63.5% (HotpotQA) to 99.2% (SciDocs). Most of that cyclicity is direct, two-way contradiction rather than longer-range nontransitivity: the construction-time mutual-drop regime (`ms1_drop_mutual`) reduces the cyclic-query rate to between 1.9%

Dataset	Regime	Cyclic (%)	After mutual (%)	FAS wt.
SciDocs	ms2	0.0	0.0	0.000
	ms1	99.2	10.8	0.074
	ms1_drop_mutual	11.7	11.7	0.002
FiQA	ms2	0.0	0.0	0.000
	ms1	98.3	30.8	0.080
	ms1_drop_mutual	30.8	30.8	0.004
HotpotQA	ms2	0.0	0.0	0.000
	ms1	63.5	1.9	0.029
	ms1_drop_mutual	1.9	1.9	0.000
BRIGHT	ms2	0.0	0.0	0.000
	ms1	92.0	22.0	0.055
	ms1_drop_mutual	22.0	22.0	0.003

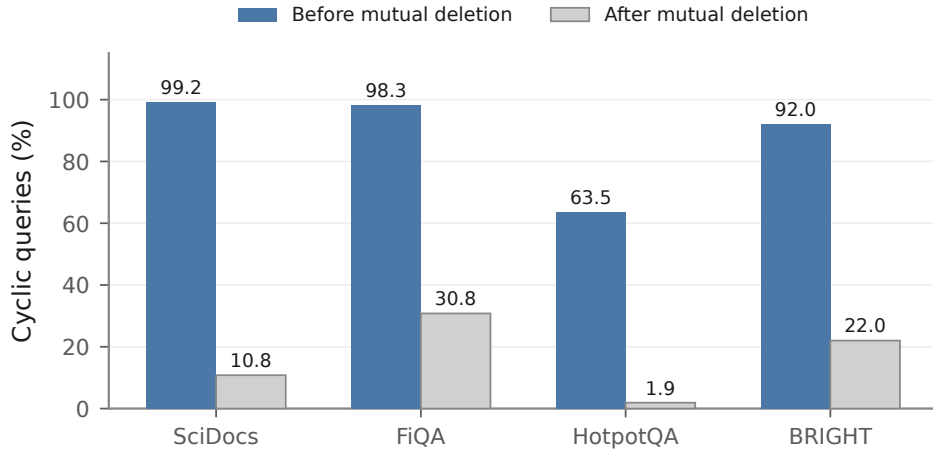


Fig. 2 RQ1 structural evidence under the canonical protocol. Top panel: cyclic-query rate, post-mutual diagnostic cyclic rate, and mean normalized feedback-arc weight by dataset and regime (After mutual: cyclic rate after deleting reciprocal edges from that regime’s aggregated graph; FAS wt.: mean feedback-arc weight removed by repair as a fraction of total graph weight). Bottom panel: active ms1 cyclic-query percentage before/after post-aggregation mutual-pair deletion. Construction, not document domain alone, decides whether repair has cyclic structure to act on.

(HotpotQA) and 30.8% (FiQA). That rate is close to, but not identical to, the post-hoc “after mutual” diagnostic on ms1 graphs: the diagnostic deletes reciprocal edges from the already-aggregated ms1 digraph, whereas ms1_drop_mutual drops contested pairs during vote construction and uses its own retention-matched aggregate threshold (Table 4). The two coincide on FiQA, HotpotQA, and BRIGHT, but differ on SciDocs (10.8% post-mutual ms1 versus 11.7% cyclic under ms1_drop_mutual; one query remains cyclic only under the construction-time drop). They therefore measure related but distinct residual longer-cycle structure (Figure 2).

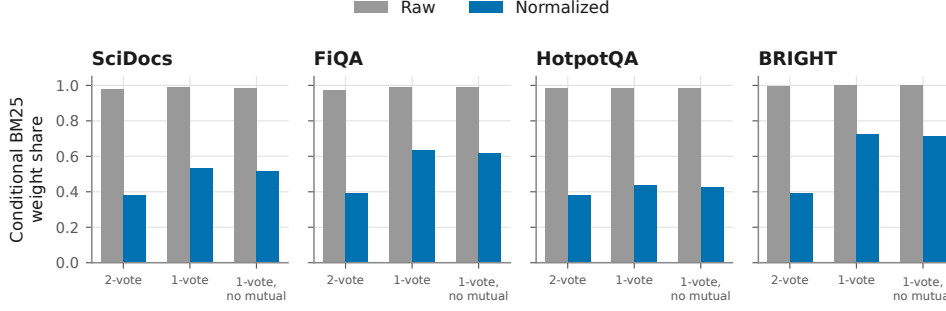


Fig. 3 RQ1 scale-dominance check: conditional BM25 edge-weight share under raw-margin ablation versus the primary normalized protocol, by dataset and regime. Raw construction is near-total BM25 domination; normalization restores a mixed multi-ranker graph before repair is studied.

Raw, unnormalized score margins are not a scale-neutral starting point for this construction (Section 3.1). Under the raw-margin ablation, BM25 contributes a mean conditional edge-weight share of 0.988 whenever it participates in an edge; under the primary normalized protocol, that share falls to 0.512. Figure 3 shows this raw-versus-normalized contrast holds across all four datasets and all three vote-construction regimes: a graph built from raw margins is close to a single-ranker graph with multi-ranker labels attached, and normalization is what restores a genuinely mixed graph.

5.2 Structural Effect of Repair

Question (RQ2).

Where graphs are cyclic, how much weight does repair remove, and does it change top- k membership when $P > k$?

Repair is structurally active precisely where Section 5.1 shows a graph has contradictory structure to remove, and inert elsewhere. Under **ms2**, repair removes nothing on any query, because no graph is cyclic. Under **ms1**, the active regime, repair removes a mean of 7.4% (SciDocs) to 8.0% (FiQA) of total graph weight; under **ms1_drop_mutual**, the fraction still to remove after mutual pairs are already gone drops to 0.2%–0.4% (Figure 2). This confirms that graph construction, not the repair algorithm, is what decides whether repair has any opportunity to act at all – a point kept separate here from whether that activity translates into a retrieval effect, which Section 5.3 addresses directly.

Repair also changes which documents are visible for evaluation once the candidate pool exceeds the metric cutoff. At $P = k$, top- k set membership cannot change by construction: the evaluated prefix is the entire pool, so repairing the graph can reorder documents but never add or remove one from the top- k set, and the measured membership-change rate is exactly 0%. Once $P > k$, pooling across all three vote-construction regimes and every larger-pool evaluation cell (Section 4.3), the mean

Table 8 RQ3 retrieval evidence under greedy repair. Panel A: confirmatory Holm-corrected repaired-vs-unrepaired nDCG families (5-pair primary method family, Section 4.8). Panel B: descriptive dataset-macro mean nDCG across the reported cells for selected anchors (graph-free methods require no repair); these macros are not a paired superiority test between CombSUM and the repaired hybrid. No cell in Panel A survives Holm correction.

Panel A. Confirmatory Holm families (paired repaired vs. unrepaired)			
Comparison family	Cells	Holm-rejected	Smallest Holm-adj. p
Active canonical (ms1)	20	0/20	0.240
Full canonical (all regimes)	60	0/60	0.720
Active larger-pool (ms1 , $P > k$ incl.)	110	0/110	0.352
Panel B. Descriptive dataset-macro mean nDCG (not paired confirmatory tests)			
Method	Mean nDCG		
CombSUM (graph-free fusion)	0.554		
Reciprocal rank fusion (graph-free)	0.546		
Copeland hybrid, repaired (ms1)	0.546		
Prior-only (graph-free)	0.545		
Borda fusion (graph-free)	0.545		

top- k membership-change rate rises to 10.6%. Repair is therefore not merely reordering an already-fixed prefix under $P > k$: it can and does change which documents are evaluated at all. Whether that structural activity is accompanied by a reliable retrieval-effectiveness gain is addressed next.

5.3 Retrieval Effectiveness

Question (RQ3).

Does greedy repair yield a Holm-surviving nDCG gain over the unrepaired graph, and do repaired methods beat graph-free fusion anchors?

Table 8 reports the Holm-corrected outcome for every prespecified repaired-versus-unrepaired nDCG comparison family under greedy repair, together with dataset-macro mean nDCG for selected graph-free anchors and the best repaired hybrid (descriptive context only).

No repaired-versus-unrepaired nDCG comparison survives Holm correction in any of the three families in Table 8 (smallest Holm-adjusted $p = 0.240$ in the active canonical family). For transparency, the SciDocs Copeland hybrid cell that yields the family’s smallest unadjusted sign-flip p (0.012) is reported here as an unadjusted cell-level result; after the prespecified Holm step-down it does not survive family-wise correction and is not confirmatory evidence. Figure 4 shows the corresponding paired bootstrap intervals for active **ms1**: point estimates are sometimes positive, but nearly every interval crosses zero.

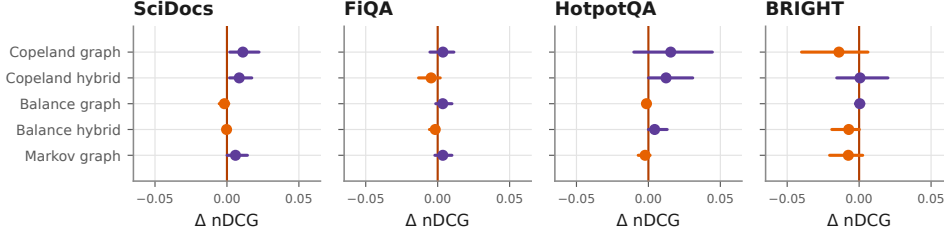


Fig. 4 RQ3 effect-size view: repaired-minus-unrepaired mean $\Delta nDCG$ with paired 95% bootstrap intervals, active **ms1**, primary five-pair family. Point estimates vary in sign; nearly every interval includes zero, consistent with the Holm null in Table 8.

Simple graph-free baselines remain competitive with the best repaired graph-dependent method in descriptive dataset-macro $nDCG$: CombSUM (0.554) exceeds every graph-dependent method’s reported macro mean, and reciprocal rank fusion and the repaired Copeland hybrid share the same reported mean to three decimals (0.546). These macros summarize performance across the reported cells; they are not a paired superiority test between CombSUM and the repaired hybrid. Under the canonical protocol, repair is not shown to beat simple score or rank fusion.

5.4 Exact Versus Heuristic Repair

Question (RQ4).

Is the RQ3 retrieval null explained by greedy suboptimality relative to a certified MWFAS edge-deletion optimum?

The comparisons in Section 5.3 use greedy repair, which does not generally reach the optimum of the feedback-arc-set objective (Equation (3)). This subsection reports what changes when repair is instead solved to certified optimality.

Of the 1,026 nominal query–regime combinations (342 usable queries $\times$ three vote-construction regimes), one BRIGHT query under **ms2** (**biology:0**) produces an empty edge set – no directed pair meets **ms2**’s min-support requirement of two rankers – and is excluded from the exact-repair study, leaving 1,025 valid exact-repair instances. Every one of those solves reached proven optimality: 1,025/1,025 query graphs, relative optimality gap 0.0, with a mean solve time of 7.4ms (median 0.25ms, maximum 236ms) against a 300-second per-query safety limit that was never approached (Section 4.6). Restricted to the 379 queries (pooled across all three vote-construction regimes) whose raw graph is actually cyclic, exact repair finds a different feedback arc set than greedy in 87.9% of them, and removes systematically less total edge weight: Table 9 and Figure 5 report the per-dataset gap. Greedy cycle-peeling is therefore a real, non-trivial overestimate of the minimum weighted feedback arc set on this evidence, not merely a theoretical possibility.

That structural gap does not translate into a retrieval-level difference. Pooled across all seven repaired-graph-dependent extraction rules evaluated in this comparison (Copeland, balance, Markov, both hybrids, topological, and priority-topological

Table 9 RQ4 structural gap on cyclic queries ($n = 379$, pooled regimes): mean feedback-arc weight removed by exact versus greedy repair. Relative reduction is from greedy to exact. Retrieval-level exact-versus-greedy and exact-versus-unrepaired Holm outcomes appear in the accompanying text (all nonsignificant after correction).

Dataset	Wt. removed (exact)	Wt. removed (greedy)	Reduction
SciDocs	6.14	8.19	25.1%
FiQA	4.51	6.37	29.2%
HotpotQA	1.55	1.80	13.4%
BRIGHT	3.07	3.95	22.3%

orderings; Section 4.7) and all five validated metrics, none of the 35 pooled metric-method cells is Holm-significant; at the finer 399-cell per-dataset-and-regime granularity, none is Holm-significant either. The largest raw, uncorrected effect in the pooled family is a mean nDCG@10 difference of -0.00223 for priority-topological ordering (raw sign-flip $p = 0.026$; Holm-adjusted $p = 0.906$).

Exact repair also leaves the unrepaired-graph comparison unchanged. With the expanded nine-method family (Table 6), 0 of 36 canonical and 0 of 56 larger-pool exact-repaired-versus-unrepaired cells are Holm-significant – the same qualitative outcome as greedy repair in Section 5.3, now under a solver-certified optimum of Equation (3). Exact repair therefore rules out greedy suboptimality with respect to the specified edge-deletion MWFAS objective, under the evaluated constructions, extraction rules, datasets, and detectable effect scale, as an explanation for a retrieval gain of that scale in Section 5.3.

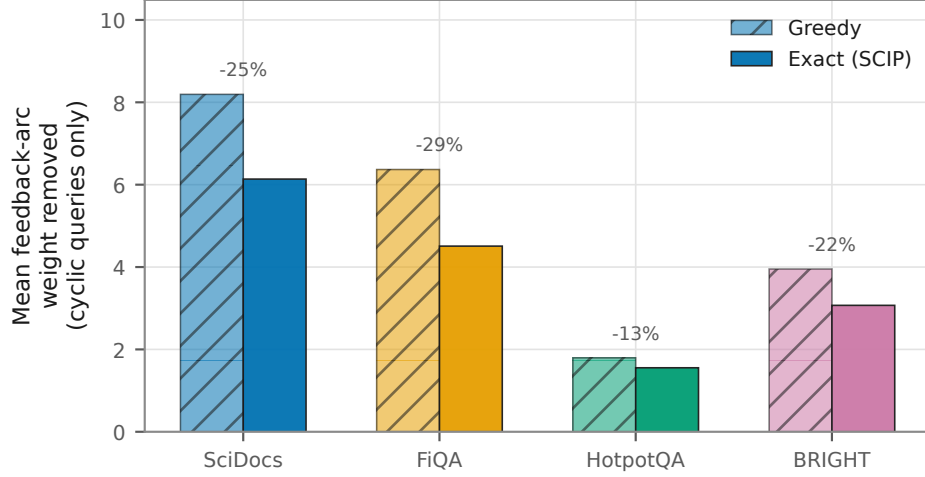
5.5 Robustness

Question.

Do pool-policy artifacts, power limits, or $P = k$ evaluation overturn the RQ3 null?

Table 10 collects four canonical robustness checks against the primary retrieval finding, each drawn from a prespecified comparison family rather than a post-hoc search.

Each check probes a distinct artifact risk for the primary null – pool advantage, power, or $P = k$ evaluation – and none overturns it. The power qualification remains: exact repair rules out greedy suboptimality with respect to the specified MWFAS objective for a gain of the detectable scale (median absolute delta 0.0036 versus Holm-adjusted median 80%-power MDE 0.0201), but a smaller general effect below that scale is not ruled out.



Exact repair consistently removes less total weight than greedy on the same graphs; both reach the same retrieval-level conclusion (no repaired-vs-unrepaired nDCG gain survives Holm correction under either repair method).

Fig. 5 RQ4 structural confirmation: mean feedback-arc weight removed by greedy versus exact repair on cyclic queries, by dataset. Exact repair removes less weight than greedy everywhere, so greedy over-removes relative to the certified optimum; this gap does not yield a Holm-surviving retrieval gain (text).

5.6 Supporting LLM Evidence

Question.

Do real LLM pairwise judgments, on a tiny query sample, contradict the score-derived RQ3 pattern?

A bounded pilot supplements the score-derived evidence above with real large-language-model pairwise judgments: six queries, four providers, and four preference-graph construction variants, analyzed at the query-cluster level ($n = 6$ independent clusters), since the pilot’s replicated rows within a query are repeated measurements, not additional independent samples. Whole-graph repair’s effect on retrieval, estimated the same way as the score-derived comparisons above, is a mean nDCG change of 0.00019 with a cluster-level 95% interval of $[-0.00072, 0.00140]$ (exact sign-flip $p = 0.875$): indistinguishable from zero, directionally consistent with the null finding in Section 5.3. Six clusters give this pilot very limited power, so this result is reported as directional and supporting only – it does not, on its own, extend any conclusion in this paper beyond the four benchmarks and construction regimes making up the main evidence package (Section 4.3).

Table 10 Robustness against the RQ3 null (canonical evidence only). Each row probes a distinct artifact risk; none overturns the Holm-corrected repaired-vs-unrepaired finding. MDE/TOST characterize detectable scale and limited equivalence, not a general proof of zero effect.

Check	Result
Baseline fairness	Testing whether the RRF-centered primary candidate pool gives a structural advantage to RRF-family baselines: 0/8 Holm-significant cells (RRF and CombSUM vs. the fixed repaired Copeland hybrid, canonical pool, all four datasets). No evidence of a pool-driven advantage.
Power and minimum detectable effect	In the active larger-pool family, median observed absolute mean delta 0.0036 vs. median Holm-adjusted 80%-power minimum detectable effect 0.0201: the study is well positioned to rule out larger effects, but many smaller ones remain below the corrected detectable scale.
Narrow equivalence	Two-one-sided-tests equivalence, active larger-pool family: 13/110 cells reject non-equivalence at a ± 0.005 margin, 32/110 at ± 0.010 , after Holm correction. Equivalence is established for a minority of cells, not the majority; the correct reading of the null result is absence of reliable positive evidence, not blanket practical equivalence.
Larger-pool ($P > k$) family	Already reported in Table 8: 0/110 Holm-significant despite a 10.6% mean top- k membership-change rate (Section 5.2), so membership change alone is not sufficient to produce a corrected retrieval gain.

6 Discussion

6.1 Main Findings

Under the canonical protocol, preference-graph repair is a real structural intervention – including under exact optimization – but not a validated general means of improving retrieval effectiveness (Sections 5.1–5.4). Structural consistency and downstream utility behave as distinct objectives here, not as the same quantity at different resolutions.

That claim is narrow. It does not say repair never helps, that acyclicity is useless, or that the null of no effect is proven. A corrected null argues against a general, reliably detectable effect at the scale this study is powered to observe (Section 5.5); smaller or dataset-specific effects remain possible. On this evidence, treating acyclicity as a surrogate for retrieval effectiveness is unsupported.

6.2 Why Structural Improvement Need Not Improve Retrieval

Section 3.5 states the formal gap: repair and nDCG use different information. Several mechanisms are consistent with genuine structural repair and a null retrieval result, without any one being isolated experimentally. Feedback-arc-set repair targets least-weight cycle-breaking edges (Section 3.4), which need not separate highly relevant from marginally relevant documents near the top of a ranking. A hybrid prior (Equation (5)) can further compress graph-side influence, consistent with graph-only and hybrid methods not diverging sharply in Section 5.3. Different extraction rules map the same repaired graph differently (Section 4.7), contributing to the mixed-sign point estimates in Figure 4. And pairwise weights and qrels are built from different evidence (Sections 3.2, 3.1), so internal consistency and relevance agreement can vary

independently. These accounts are consistent with the design and observed pattern, not causal claims established by a dedicated test.

6.3 Exact Repair as a Diagnostic Control

A heuristic-only study leaves open graph-objective suboptimality as an alternative explanation. Section 5.4 addresses that alternative for this evidence only with respect to the specified edge-deletion MWFAS objective on solver-certified instances: exact repair removes less edge weight than greedy (typical over-removal by the heuristic) yet yields no Holm-surviving retrieval gain against the unrepaired graph under the evaluated constructions, extraction rules, datasets, and detectable effect scale. Exact SCIP repair is therefore a methodological identification control (Section 3.4), not a novel solver, not a proof that no graph intervention can help retrieval, and not a recommendation to run exact MWFAS at production scale.

6.4 Relation to Prior Literature

The finding that repair is structurally active but not reliably retrieval-improving sits beside the closest lines in Table 1. Pairwise LLM ranking and graph aggregation ask whether pairwise evidence or inconsistency mitigation improves ranking relative to alternatives [30, 37, 54]; acyclic preference methods report gains on LLM evaluation, selection, or RLAIIF objectives [19, 28]. Those results are compatible with a null repair effect for document-retrieval nDCG under the present construction-factor protocol. The earlier preprint comparison remains in Table 2.

6.5 Practical Implications

The reusable methodological lesson is the factorized audit itself: treat construction, structural intervention, extraction, and utility as separate evidence streams, and use exact MWFAS on a bounded subset as an identification control when a heuristic null must be stress-tested. Practitioners should take three points from this evidence: structural metrics diagnose construction quality but are not validated nDCG surrogates here; a repair stage is worth evaluating—with unrepaired and graph-free fusion controls—when a pipeline already builds a preference graph and treats acyclicity as a quality signal; and exact repair is a diagnostic tool for graph-objective suboptimality on a representative subset, not a full-scale deployment recommendation. Report construction, extraction, and cutoff with any repair claim, and keep structural and utility metrics as separate streams (Section 5.1). These are audit practices, not a validated deployment rule or a production cost claim. Predicting per-query whether repair will help remains future work.

6.6 Implications for LLM-Based Ranking

LLM pairwise judgments can induce cycles for the same structural reasons as score-derived votes (Section 3.2). The structural-versus-utility distinction therefore applies in principle, but this paper’s LLM evidence is only the six-query pilot in Section 5.6

and does not support broad claims about modern LLM rerankers. Extending the controlled comparison remains future work.

7 Limitations and Threats to Validity

7.1 Internal Validity

The pipeline (Section 4.2) makes sequential design choices – normalization, vote thresholds, pooling, and hybrid weight α – that could interact with the repair effect. Two controls mitigate this: retention-matched thresholding (Section 4.5) ties normalized permissiveness to the raw-margin baseline without using qrels, and every retrieval comparison holds construction, pooling, and extraction fixed while varying only repair (Section 5.3). Threshold calibration on the same usable query list remains a transductive protocol choice: a held-out calibration split was not used and is a natural future robustness check. What also remains open is α -dependence beyond the reported sweep (Section 4.7). Paired sign-flip with Holm correction (Section 4.10) assumes exchangeable paired differences within a family; query-level pairing supports that assumption within a cell, but the same queries recur across regimes and pool sizes. Holm therefore controls cell-wise family-wise error among the listed tests without a hierarchical model of cross-cell dependence; multilevel modeling of that structure is future work, not a claim of the present analysis.

7.2 Construct Validity

Acyclicity is one narrow notion of structural consistency (Section 3.3); the reported diagnostics (cyclic-query rate, removed feedback-arc weight, top- k membership change) measure graph-internal coherence, not evidential correctness. nDCG is the primary utility construct; MRR and MAP are secondary diagnostics only (Section 4.9). Pairwise vote weight encodes relative-order confidence, not absolute relevance (Section 3.2); the two are related through shared documents but are not the same measurement (Section 3.5).

7.3 External Validity

The main evidence is offline multi-ranker reranking on four public benchmarks with score-derived graphs (Sections 4.3, 4.4), not web-scale or conversational retrieval. Pools are shallow (35–50) and RRF-centered; base rankers are BM25/TF-IDF/MiniLM as controlled score sources, not a leaderboard. The real-LLM pilot is the six-query addendum of Section 5.6. Cross-encoders, stronger dense/hybrid stacks, listwise/tournament LLM rankers [6, 36], and neural graph rankers are incomparable as unrepaired-versus-repaired cells and remain future work. Transfer may also fail under relevance-aware weights, edge-reversal or soft repair, sparse comparisons, deeper pools, interactive retrieval, or effects below the corrected detectable scale.

7.4 Statistical Conclusion Validity

Confirmatory claims use Holm-corrected paired inference over reported families (Section 4.10). No Holm rejection is not proof of no effect. The study can rule out a general effect at the scale of the observed point estimates relative to the corrected MDE (Section 5.5), but smaller effects remain possible. TOST supports narrow equivalence only for cells that meet the prespecified margin; equivalence in a minority of cells does not establish blanket practical equivalence. BH is reported as a less conservative robustness comparison, not a replacement for confirmatory Holm. The real-LLM pilot uses cluster-level ($n = 6$) inference (Section 4.10) and is not confirmatory.

7.5 Computational Limitations

Exact MWFAS reaches certified optimality quickly at the query-graph sizes studied here (Section 5.4); larger graphs and deeper candidate pools are not evaluated, so scalability conclusions stop at the reported solve-time distribution. Exact repair remains the diagnostic control of Section 6.3. Greedy cycle-peeling remains the practical choice for larger graphs because exact MWFAS is NP-hard in general (Section 2.3).

8 Conclusion

This study establishes, under the reported protocol, that construction choices decide whether preference graphs are cyclic; that greedy and exact repair both restore acyclicity where structure exists, with exact repair removing less weight; and that neither yields a Holm-surviving repaired-versus-unrepaired nDCG gain in the canonical, larger-pool, or exact-repair families – even when larger-pool repair changes top- k membership. Exact repair rules out greedy suboptimality with respect to the specified edge-deletion MWFAS objective on solver-certified instances under the evaluated constructions, extraction rules, datasets, and detectable effect scale, without revealing a hidden retrieval gain of that scale. Equivalence evidence remains limited to a minority of TOST cells; smaller effects below the corrected minimum-detectable scale remain possible.

The study does not establish that repair never helps, that repaired and unrepaired rankings are universally equivalent, or that exact MWFAS should be run at production scale. Its methodological usefulness is the factorized audit: report construction, structural diagnostics, extraction, and downstream metrics separately; keep an unrepaired baseline; and use exact repair as a bounded identification control when a weak downstream result must be stress-tested (Section 6.5). Future preference-graph and LLM-judge studies should adopt that separation rather than treating acyclicity as a proxy for task utility.

9 Data Availability and Reproducibility

The public benchmarks used in this study are available from their original sources: SciDocs, FiQA, HotpotQA, and BRIGHT are cited in Section 4.3. The code, fixed query lists, processed intermediates, figure data, and scripts required to reproduce

the reported tables and figures are available at <https://github.com/SoroushVahidi/consistency-aware-llm-rankin>. The main-paper score-derived results are regenerated from stored ranker-score and relevance-judgment artifacts; no paid provider call is required to recompute those statistics. The bounded real-large-language-model pilot reported in Section 5.6 used paid provider APIs (Microsoft Azure / OpenAI-compatible, Google Gemini, Cohere, and Fireworks AI) and is supported in the repository by sanitized derived artifacts and usage summaries only. Raw provider request and response payloads are not included, because they may contain prompt/completion content and operational details excluded by the project’s artifact policy.

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Statements and Declarations

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Competing interests

The author declares that there are no competing interests.

Ethics approval

Not applicable. This study uses only publicly available benchmark datasets and stored, derived score files; it involves no human subjects, no personal or identifiable data, and no new data collection from human participants.

Consent to participate

Not applicable; no human participants were involved in this study.

Consent for publication

Not applicable; no individual person’s data are presented in this manuscript.

Data availability

The datasets used in this study are publicly available from their original sources and are cited in the manuscript. No new benchmark dataset was generated. Compact processed artifacts used to generate the reported results are available in the public repository linked under Code availability. Raw provider request and response payloads from the bounded large-language-model pilot are not included in the repository, because they may contain prompt/completion content and operational details excluded by the project’s artifact policy. See also Section 9.

Code availability

The code, fixed query lists, processed intermediates, figure data, and scripts required to reproduce the reported tables and figures are available at <https://github.com/SorushVahidi/consistency-aware-llm-rankin>.

Materials availability

Not applicable. This study does not generate new physical materials.

Authors’ contributions

This is single-author work. Soroush Vahidi conceived the study, designed and implemented the experimental pipeline, ran the experiments, analyzed and verified the results, prepared the figures and tables, and wrote the manuscript.

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